GATE MOCK PAPER - 2 2026
Data Science & Artificial Intelligence (DA)

General Aptitude
Q.1 – Q.5 Carry ONE mark Each

Q.1	If "$\rightarrow$" denotes increasing order of intensity, then Cold $\rightarrow$ Chill $\rightarrow$ Frost is analogous to: Smile $\rightarrow$ ________ $\rightarrow$ Laughter
(A)	Grin
(B)	Smirk
(C)	Giggle
(D)	Frown

Q.2	Read the passage carefully. Some scientific theories are initially rejected because they contradict prevailing beliefs. Over time, experimental evidence may lead to their acceptance. Which statement must be TRUE?
(A)	All new theories are correct
(B)	Prevailing beliefs are always wrong
(C)	Experimental evidence can change acceptance of theories
(D)	Rejected theories never gain acceptance

Q.3	Two cards are drawn randomly from a standard deck of 52 cards without replacement. What is the probability that both cards are kings?
(A)	$\frac{1}{221}$
(B)	$\frac{1}{169}$
(C)	$\frac{6}{1326}$
(D)	$\frac{1}{52}$

Q.4	A circle of radius 3 cm is inscribed in a square. The area of the region inside the square but outside the circle is:
(A)	$36 - 9\pi$
(B)	9π
(C)	18π
(D)	$12 - 9\pi$

Q.5	The average of 8 numbers is 15. If one number 23 is replaced by 7, the new average is:
(A)	13
(B)	14
(C)	15
(D)	16

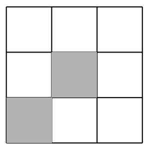
Q.6 – Q.10 Carry TWO marks Each

Q.6	Let $g(x) = x^2 + 6x + 5$. Which statements are correct?
(A)	The parabola opens upward
(B)	Minimum occurs at $x = -3$
(C)	$g(x) = 0$ has real roots
(D)	The minimum value is -4

Q.7	Events A and B satisfy $P(A) = 0.7$, $P(B) = 0.6$, $P(A \cap B) = 0.42$. Which are TRUE?
(A)	A and B are independent
(B)	$P(A\|B) = 0.7$
(C)	$P(A \cup B) = 0.88$
(D)	$P(B\|A) = 0.6$

Q.8	Consider the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$. Which are correct?
(A)	The series converges
(B)	Partial sum is $1 - \frac{1}{n+1}$
(C)	The sum equals 1
(D)	The series diverges

Q.9	A 4×4 grid initially has only corner squares shaded. A square becomes shaded if it has exactly two shaded neighbors. After stabilization, how many squares are shaded?
(A)	4
(B)	8
(C)	12
(D)	0

Q.10	Which option completes the pattern shown below? 
(A)	Diagonal continuation
(B)	Horizontal fill
(C)	Vertical fill
(D)	Random fill

Q.11 – Q.35 Carry ONE mark Each

Q.11	If two random variables are independent, which statement is ALWAYS true?
(A)	Zero covariance
(B)	Same distribution
(C)	Same variance
(D)	Linear relation

Q.12	Three fair dice are thrown independently. Let A be the event that the sum is at least 15 and B be the event that all dice show distinct numbers. What is $P(A \cap B)$?
(A)	0
(B)	$\frac{1}{36}$
(C)	$\frac{1}{18}$
(D)	$\frac{5}{36}$

Q.13	A matrix has eigenvalues $1, -2, 3$. Which statement is TRUE?
(A)	Determinant is -6
(B)	Trace is 2
(C)	Rank is 3
(D)	Matrix is singular

Q.14	During BFS on an unweighted graph, which edge can NEVER occur?
(A)	Tree edge
(B)	Back edge
(C)	Cross edge
(D)	Forward edge

Q.15	If $f'(x_0) = 0$ and $f''(x_0) < 0$, then x_0 is a
(A)	Point of inflection
(B)	Local minimum
(C)	Local maximum
(D)	Saddle point

Q.16	Which data structure supports amortized $O(1)$ insertion at both ends?
(A)	Stack
(B)	Queue
(C)	Deque
(D)	Priority queue

Q.17	Which kernel allows SVM to create nonlinear decision boundaries?
(A)	Linear
(B)	Polynomial
(C)	RBF
(D)	All of the above

Q.18	PCA is primarily used for
(A)	Classification
(B)	Dimensionality reduction
(C)	Clustering
(D)	Regression

Q.19	In k-means clustering, increasing k generally
(A)	Increases SSE
(B)	Decreases SSE
(C)	Keeps SSE constant
(D)	Makes clustering impossible

Q.20	Which regularization corresponds to Laplace prior?
(A)	Ridge
(B)	Lasso
(C)	Elastic Net
(D)	None

Q.21	The recurrence $T(n) = 2T(n/2) + n$ solves to
(A)	$O(n)$
(B)	$O(n \log n)$
(C)	$O(n^2)$
(D)	$O(\log n)$

Q.22	Curse of dimensionality mainly affects
(A)	k-NN
(B)	Linear regression
(C)	Naïve Bayes
(D)	Decision trees

Q.23	Which logical formula is a contradiction?
(A)	$p \vee \neg p$
(B)	$p \wedge \neg p$
(C)	$p \to p$
(D)	$\neg(p \wedge q)$

Q.24	If $X \sim U(0,1)$, then $E[X^2]$ equals
(A)	$\frac{1}{2}$
(B)	$\frac{1}{3}$
(C)	$\frac{1}{4}$
(D)	$\frac{2}{3}$

Q.25	Which problem is NP-complete?
(A)	Vertex cover
(B)	Minimum spanning tree
(C)	Shortest path
(D)	Bipartite matching

Q.36 – Q.65 Carry TWO marks Each

Q.36	$X_1, \ldots, X_{100}$ are i.i.d. Poisson(3). Using CLT, approximate $P(260 \leq \sum X_i \leq 340)$.
(A)	$\Phi(1) - \Phi(-1)$
(B)	$\Phi(2) - \Phi(-2)$
(C)	$\Phi(0.5) - \Phi(-0.5)$
(D)	$\Phi(3) - \Phi(-3)$

Q.37	Which statements about sample mean are correct?
(A)	Unbiased estimator
(B)	Variance decreases with n
(C)	Consistent estimator
(D)	Variance equals population variance

Q.38	In Bayesian Network $A \rightarrow B \leftarrow C$, which is TRUE?
(A)	$A \perp C$
(B)	A and C become dependent given B
(C)	B blocks information
(D)	All

Q.39	Which SVD property is correct?
(A)	Rank equals number of nonzero singular values
(B)	Singular values may be negative
(C)	U and V are non-orthogonal
(D)	Always square

Q.40	Which are TRUE for orthogonal matrices?
(A)	$A^T A = I$
(B)	$\lvert \det(A) \rvert = 1$
(C)	Preserves length
(D)	Always symmetric

Q.41	Consider the function $$f(x) = x^4 - 4x^3 + 6x^2 - 4x + 1.$$ Which of the following are TRUE?
(A)	$x = 1$ is a global minimum
(B)	$f(x)$ is convex for all x
(C)	$f'(1) = 0$
(D)	$f(x) = (x-1)^4$

Q.42	Which of the following statements are TRUE in relational databases?
(A)	Every superkey contains a candidate key
(B)	A candidate key may have redundant attributes
(C)	BCNF is stricter than 3NF
(D)	3NF always eliminates update anomalies

Q.43	Which of the following statements are TRUE about decision trees?
(A)	Gini impurity is faster to compute than entropy
(B)	Information gain prefers balanced splits
(C)	Trees are prone to overfitting
(D)	Pruning reduces bias

Q.44	Which SQL queries are valid?
(A)	SELECT dept, COUNT(*) FROM Emp GROUP BY dept;
(B)	SELECT dept, salary FROM Emp GROUP BY dept;
(C)	SELECT COUNT(DISTINCT dept) FROM Emp;
(D)	SELECT dept FROM Emp HAVING COUNT(*)¿2;

Q.45	Which problems can be solved optimally using greedy strategy?
(A)	Fractional knapsack
(B)	Activity selection
(C)	Huffman coding
(D)	0/1 knapsack

Q.46	Which properties of heuristic $h(n)$ ensure optimality of A* search?
(A)	Admissibility
(B)	Consistency
(C)	Overestimation
(D)	Non-negativity

Q.47	Which statements are TRUE about Bellman–Ford algorithm?
(A)	Handles negative edges
(B)	Detects negative cycles
(C)	Faster than Dijkstra in all cases
(D)	Time complexity is $O(VE)$

Q.48	A relation has 6 attributes and two candidate keys of size 3 each. How many prime attributes does it have? ________________

Q.49	A connected graph with n vertices is a tree. Minimum number of edges to be added so that the graph becomes 2-edge-connected is ________________

Q.50	Let X, Y be independent random variables. Which are TRUE?
(A)	$E[XY] = E[X]E[Y]$
(B)	$Cov(X, Y) = 0$
(C)	They must be Gaussian
(D)	They are uncorrelated

Q.51	Let A be an orthogonal matrix with $\det(A) = -1$. Which statements are TRUE?
(A)	$A^{-1} = A^T$
(B)	Represents a reflection
(C)	Preserves vector norms
(D)	Must be symmetric

Q.52	Which statements about normalization are TRUE?
(A)	BCNF implies 3NF
(B)	3NF implies BCNF
(C)	BCNF may lose dependencies
(D)	3NF always preserves dependencies

Q.53	Which properties hold for B^+-trees?
(A)	All records at leaves
(B)	Leaves are linked
(C)	Height grows logarithmically
(D)	Internal nodes store full records

Q.54	For a graph with negative edges but no negative cycle:
(A)	Dijkstra fails
(B)	Bellman–Ford works
(C)	Negative edge implies negative cycle
(D)	Bellman–Ford detects cycles

Q.55	A deck of 52 cards is dealt equally among 4 players. Probability that each gets exactly one ace is
(A)	$\frac{4!\binom{48}{12}^4}{\binom{52}{13}^4}$
(B)	$\frac{\binom{4}{1}^4}{\binom{52}{13}}$
(C)	$\frac{1}{13^4}$
(D)	$\frac{4!}{52!}$

Q.56	Let X_n be Bernoulli with $P(X_n = 1) = 1/n$. Which modes of convergence to 0 hold?
(A)	In probability
(B)	In mean
(C)	Almost surely
(D)	In distribution

Q.57	Which are advantages of Gini impurity over entropy?
(A)	Faster computation
(B)	Less sensitive to class imbalance
(C)	Produces simpler trees
(D)	Always more accurate

Q.58	A matrix has eigenvalues $2, 4, 8$. Find trace(A^{-1}). ____________________

Q.59	Which statement about alpha–beta pruning is TRUE?
(A)	Changes minimax value
(B)	Prunes irrelevant branches
(C)	Always reduces complexity
(D)	Applies only to stochastic games

Q.60	For $f(x) = \begin{cases} \dfrac{\sin(ax)}{x}, & x \neq 0 \\ b, & x = 0 \end{cases}$ Which are TRUE?
(A)	Continuous at 0 if $b = a$
(B)	Differentiable at 0 if $a \neq 0$
(C)	$f'(0) = 0$ if $a = 1$
(D)	Not differentiable if $a \neq 0$

Q.61	A binary min-heap initially has 511 elements. Maximum number of comparisons during one Insert followed by Extract-Min is ____________________

Q.62	A binary heap has 20 elements. Insert 4 elements and delete 3 minimum elements. Final heap size is ____________________

Q.63	A full binary game tree of depth 5 is searched using minimax without pruning. Number of leaf nodes evaluated is ____________________

Q.64	Relation $R(A, B, C, D)$ has FDs: $A \to B$, $B \to C$, $C \to D$. Number of candidate keys is ____________________

Q.65	A B^{+}-tree of order 5 indexes 3000 keys. Minimum possible height is ____________________